



THE
NIGHTINGALE
MAPPING

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Revealing the hidden mathematical structure of sound

Official Project Document

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A premium summary of the project roadmap, foundation layer, and milestone system for building a bidirectional translation layer between sound and mathematics.

Only real, locally verified results from the repository will be shared publicly.
Project grounded in first principles - built for discovery.

Core Objectives

1. Structure-Preserving Bidirectional Layer (Foundation)

Real audio -> mathematical ratio ladder -> regenerated audio that preserves core structure under pitch shift, time scaling, harmonic density, and composition.

2. Reproducibility & Verification (Truth Layer)

Every result must be fully reproducible from the public repository with one command. Local repo = single source of truth.

3. Optimization Engine (Improvement Layer)

A genuine self-improving loop that evolves parameters using real, non-saturated metrics and delivers measurable gains in structural invariance.

4. Temporal & Rhythm Layer

Extend the ladder into the time domain with reliable IOI extraction, tempo invariance, and bidirectional rhythm mapping.

5. Discovery Layer - New Mathematics from Nature

Apply the full system to nightingale song to uncover non-obvious symmetries, recursive structures, and new mathematical objects.

6. Scaling, Demonstration & Readiness

Create compelling generative examples, robust documentation, and an investment-ready pipeline.

Current Progress - April 2026

We are currently focused on **Section 1 (Foundation)** and **Section 3 (Optimization Engine)**.

- Bidirectional test pipeline and self-improvement loop are operational
- Next priorities: locking real pitch invariance (M1.2) and building a meaningful optimization metric (M3.1)

Full Milestones Checklist (v1.1)

Section 1 - Structure-Preserving Bidirectional Layer

M1.1 - Synthetic Round-Trip Lock

100% exact Fraction match on ≥ 5 canonical test sequences

M1.2 - Real Audio Pitch Invariance

$\geq 7/8$ ratios recovered within 15 cents tolerance on orchestra + rock clips (pitch shift)

M1.3 - Real Audio Time Stretch Invariance

$\geq 7/8$ ratios preserved under 1.5x time stretch

M1.4 - Combined Transformation Stability

≤ 1 mismatch per 8-note sequence under sequential transforms

Section 2 - Reproducibility & Verification

M2.1 - One-Command Baseline

Repository baseline runs cleanly from a fresh setup

M2.2 - Run-to-Run Stability

≤ 1 note difference across 5 runs

M2.3 - Fresh Environment Test

Consistent results on a new Replit or machine

Section 3 - Optimization Engine

M3.1 - Real Invariance Metric

Non-saturated metric that varies meaningfully with config

M3.2 - Config-Driven Gain

Best config improves average invariance by $\geq 8\%$ over baseline

M3.3 - Generalization Test

≥ 0.65 invariance on held-out nightingale clip

Section 4 - Temporal & Rhythm Layer

M4.1 - Reliable IOI Extraction

$< 10\%$ false onsets with consistent categorical ratios

M4.2 - Rhythm Invariance

$\geq 80\%$ preservation of relative IOI ratios under time stretch

M4.3 - Bidirectional Rhythm Mapping

$\geq 75\%$ structure match after encode-regenerate-decode

Section 5 - Discovery Layer

M5.1 - Non-Trivial Pattern Detection

Identify ≥ 3 repeatable non-obvious ratio/symmetry patterns

M5.2 - Cross-Clip Consistency

≥ 2 shared structural motifs across multiple nightingale phrases

Section 6 - Scaling, Demonstration & Readiness

M6.1 - Structure-Preserving Generation

Generative examples retain structural integrity under transformation

M6.2 - Public Demo Package

Audio examples + ratios + clear instructions

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